

Complex Numbers



Powers of i

1 Evaluate each power of i , using $i^2 = -1$ and the fact that powers of i repeat every 4:

a i^3

b i^4

c i^7

d i^{22}

2 Evaluate i^{101} .

Addition and subtraction

3 Let $z_1 = 3 + 4i$ and $z_2 = -2 + 5i$. Find $z_1 + z_2$.

4 Let $z_1 = 2 - 3i$ and $z_2 = -1 - 4i$. Find $z_1 + z_2$.

5 Find $(7 + 2i) - (3 - 5i)$.

6 Find $(4 - i) - (-2 + 6i)$.

Multiplication

7 Multiply $(2 + 3i)(1 - 4i)$.

8 Multiply $(-1 + 2i)(3 + i)$.

9 Multiply $(5 - 2i)(5 + 2i)$, and explain why the result has no imaginary part.

Conjugate and modulus

10 For $z = 6 - 8i$, find $\bar{z}$ and $|z|$.

11 For $z = -5 + 12i$, find $\bar{z}$ and $|z|$.

12 Show that $z + \bar{z}$ is always real for $z = a + bi$.

Division

- 13 Simplify $\frac{3+4i}{1-2i}$ by multiplying top and bottom by the conjugate of the denominator.
- 14 Simplify $\frac{1+i}{1-i}$.
- 15 Simplify $\frac{7-i}{3+2i}$.
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Quadratic equations with complex roots

- 16 Solve $x^2 + 9 = 0$.
- 17 Solve $x^2 - 6x + 13 = 0$ using the quadratic formula.
- 18 Solve $2x^2 + 2x + 5 = 0$ using the quadratic formula.